THE SONIC BOLLART

Tim Gilchrist
The Perth Artifactory
8/16 Guthrie Street
Osborne Park
Western Australia

Meg Travers
Western Australian Academy of Performing Arts
Edith Cowan University
Perth, Western Australia, Australia

Tara Surry
The Perth Artifactory
8/16 Guthrie Street
Osborne Park
Western Australia

SKoT McDonald
The Perth Artifactory
8/16 Guthrie Street
Osborne Park
Western Australia

Morgan Strong
Project Manager, Digital Strategy
Department of Science, Innovation and Information Technology
Queensland Government

Craig Wale
The Perth Artifactory
8/16 Guthrie Street
Osborne Park
Western Australia

ABSTRACT

BollART is a concept developed during the Perth International Arts Festival's 'art-hack'. It was conceived during the art-hack event, and its purpose is to provide a digital art platform in public spaces. The broad concept - which was rapidly conceived under the constraints of a hack event - was to take a mundane item of urban furniture, and repurpose it as a general-purpose digital artwork platform. The Sonic BollART is a reference implementation/prototype developed as a proof of concept for our idea.

1. INTRODUCTION

The Sonic BollART is an interactive exhibit, conceived and prototyped as part of Australia's inaugural 'art hack', Hack The Festival held during the Perth International Arts Festival 2015 (Perth International Arts Festival 2015). The brief for Hack The Festival was to "have 24 hours to create a digitally interactive artwork", followed by a week to turn the idea into a functioning prototype. Over the course of the event, our team went from brainstorming ideas, to a functioning reference implementation, to winning the competition.

Our project was conceived as a general-purpose urban digital art platform, transforming a mundane and practical real world object - the bollard - into a vessel which artists, musicians, students and hackers/makers can use as a virtual stage to contain their digital creations. Reflecting the interests of our group as well as current local arts issues, our prototype was both an electronic musical instrument, and a WiFi dead-drop for distributing local music.



Figure 1 The Sonic BollART Prototype.

2. THE ART HACK

Hack the Festival began with an information session, where ideas could be pitched to other attendees and teams formed. Our team - Team Artifactory - formed around the idea of modifying robot vacuum cleaners to rove autonomously around cultural spaces armed with

tablet computers, engaging with the public and provoking them in topical discussions.

The idea of argumentative Roombas was swiftly rejected, as we felt that we wouldn't be able to complete a prototype which sufficiently embodied our ideas, nor deal with potential safety issues[1], under the tight time constraints of the event. Further brainstorming led us to the idea that we could fit a Theremin inside bollard. As these objects are often constructed of metal, the cladding could be connected to the Theremin to form the pitch antenna part of the circuit.

Hack events thrive on rapidly evolving ideas, deciding if they'll fit, if they are achievable, discarding them and changing direction as needs be. Our path from vague idea to Bollard to BollART saw us consider the urban furniture available to us. Bins, lights, fences, rails, as well as existing pieces of urban art were considered as the target of our hack. The bollard was ultimately chosen, as they are commonly deployed in situations where people pass them closely enough to interact.

Over the course of the hack, we developed this basic idea to include a wireless music dead-drop, where local bands could upload and share their music. The dead drop would be disconnected from the Internet, so the public would need to be physically present to be involved.

By the end of the hack day proper, we had the beginnings of a physical object. This was refined over the next few evenings until we had a functioning prototype.

3. BOLLART CONCEPT

We conceived The Sonic BollART as a sound installation, which addressed the problem of accessibility within an urban space to those with visual impairment.

The bollard represents a significant opportunity for public art, and a serious barrier to access. We started by thinking "how can we make a bollard more accessible"?

Bollards do not only restrict access, they create barriers. People with vision impairment cannot find bollards easily. Sound addresses this issue. If a bollard can make sound in relation to proximity it makes it more accessible to everyone. And if we can turn them into a musical instrument, then we make this mundane object both playful and accessible. (Strong 2015)

As a mechanism for disseminating local music, a wireless music dead drop was added. Several members of our team are involved in the local music scene, and there has been much discussion across Perth recently on the lack of music venues willing to take live bands into their entertainment roster - a problem not unique to Western Australia in times where venues and new residential buildings are in close proximity. They also noted that finding new audiences for their work - and engaging them - can be difficult. Having a device where local musicians can upload their music for the public to access it seemed like an ideal second purpose for our project.

Our plans also extended to include "Easter Eggs" - secret functionality often present in software, DVDs and Albums (Wikipedia Contributors 2015). Ideas such as playing particular note sequences on the Theremin unlocked "Disco Mode" light shows and playing tracks from the music dead-drop were raised.

With these core ideas in place, we pitched The BollART as a generic art installation platform, where bollard-laden public spaces could be used as ready-made digital sculpture gardens. By providing basic infrastructure (physical frame and a source of electricity and network access), other artists, musicians, makers and students could use the BollART shells as platforms to create their own digital art.

Just like "cow parade" events involving hundreds of custom painted cows placed in cities across the world, the concept of BollART has similar potential. However, the BollART concept has one fundamental difference: bollards are standard infrastructure in the urban environment, and by converting them into BollART, they are creatively coaxed far beyond its basic raison d'être to delight and engage passersby. Whilst also public art, each BollART is part of the real, functional and normally unsung (and unsinging) working fabric of the City.

4. REFERENCE IMPLEMENTATION

Our prototype consisted of:

- Moog Theremini Theremin controlling addressable RGB LED light strips via an Arduino Microcontroller board;
- Raspberry Pi computer running Linux and the PirateBox web application for the dead-drop
- A laser cut MDF frame for housing the components.

4.1. Physical Construction

We built our bollard chassis using the resources of local Makerspace - The Perth Artifactory[2]. The frame was designed to house and support the various components and power supplies, be simple enough to construct the time we had, and sturdy enough to deploy as a functioning prototype. Once designed, the wooden pieces were laser cut and brought back to the hack venue for final assembly.

[1] Low to the ground, free-roving robots in a public space posing a trip hazard to the unsuspecting public, for example.

[2] http://artifactory.org.au/

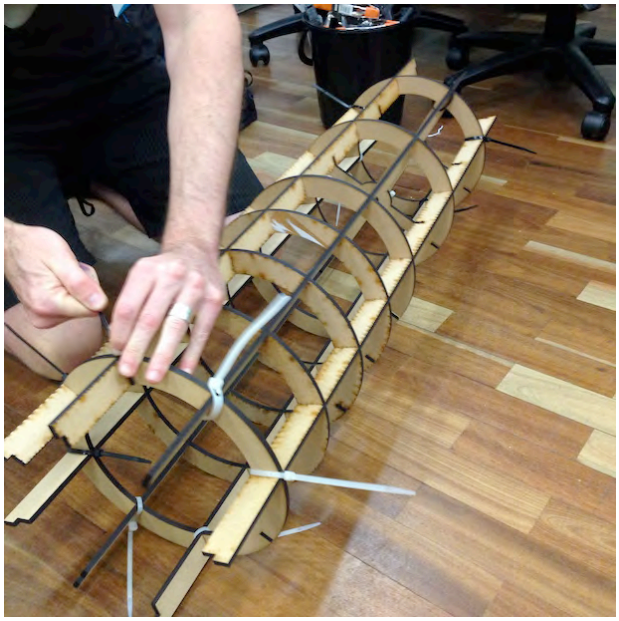


Figure 2 The laser cut BollART Frame under construction.

4.2. Theremin

A Moog Theremini was chosen as the Theremin we would use, as our team had ready access to one and they are more reliable than the kits available from places like Jaycar and other local hobbyist electronics shops. The pitch antenna, which normally stands vertically from the case was removed, and a length of copper pipe was mounted centrally in the upper section of the frame, and wired to the Theremin circuit.

The Theremini was suspended inside the frame by its volume antenna and plastic cable ties. This arrangement proved a less than ideal choice, as both the antennas were in close proximity, making setup of the instrument difficult - little use was in fact made of the volume antenna as 'players' would be located too far from the Theremin to make it functional, however the Theremini setup requires that the volume antenna be tuned as well as the now much larger pitch antenna. Performance was deemed adequate for a prototype, so no further modifications were made.

4.3. Wireless music dead-drop

The wireless music dead drop was implemented on a low-cost Raspberry Pi computer running PirateBox[1]. PirateBox is an open-source (free software) Linux distribution, which comes preconfigured with software to run an anonymous file-sharing platform over a private WiFi network.

Although anonymity was not our primary concern, the software was otherwise fit for our purposes. It provided a mechanism for uploading and downloading of files. Android, MacOS and iOS devices were tested, and all were capable of playing music directly from the dead-drop.



Figure 3 Raspberry Pi configured with PirateBox

4.4. Easter Eggs

The "Easter Egg" concept was implemented in our prototype as a software controlled light show using the CV pitch output of the theremin. We decided to make this an obvious function instead of a hidden feature for intrepid users to discover - again due to the tight time constraints of the hack.

Lighting was achieved using addressable LED (Light Emitting Diode) strip. This is a flexible, weatherproof printed circuit board with LED modules and circuitry embedded into it. This strip can be controlled directly from an Arduino microcontroller and several other small, low cost computing platforms.

[1] http://piratebox.cc/



Figure 4 Addressable LED strip installed on the BollART prototype.

The LEDs were arranged as a seven level bar graph pattern, with higher pitches resulting in more LEDs being illuminated. Seven levels were chosen, as that number of LED modules fit comfortably on the bollard frame. The effect we were targeting was that of a 1980's style LED VU meter.

The Moog Theremini provides a configurable CV output. Zero to five volts was chosen for the output range, and fed to an Arduino Uno microcontroller board via one of the analog to digital converter (ADC) pins. This gave us an integer value which varied with pitch between 0 and 1023, for 10 bits of resolution at roughly 4.9mV per increment. Software was written to take the ADC output value, and process it for display on the addressable (software controllable) LED strip.

The 10-bit input values from the theremin were mapped (bucketed) to seven levels, with the remainder used to module the brightness of the final segment in real time. Each segment is fully lit for a value of approximately 146 of the total 1023 range. For example, if the Theremin's pitch output gave a number of 400, the first two segments would be illuminated full brightness, with the third segment at roughly 73% brightness.

LED 1	LED 2	LED 3	LEDs 4..7
0 to 146	147 to 292	293 to 400	N/A
100%	100%	73%	0%

Table 1 Bucketing of an ADC value of 400 for each LED segment.

The net result of this is that when the theremin plays a pitch, the LEDs light up to match.

5. PRODUCTION DESIGN

The ability for the artwork to be deployable in a public space was a key criterion for judging, so we were keen to address this. To expand on our proof of concept, a design for a robust "BollART" platform was produced. A production-ready BollART would be made from concrete, fibreglass and/or metal. 3D renderings were produced for use in our presentation, and a logo was designed.

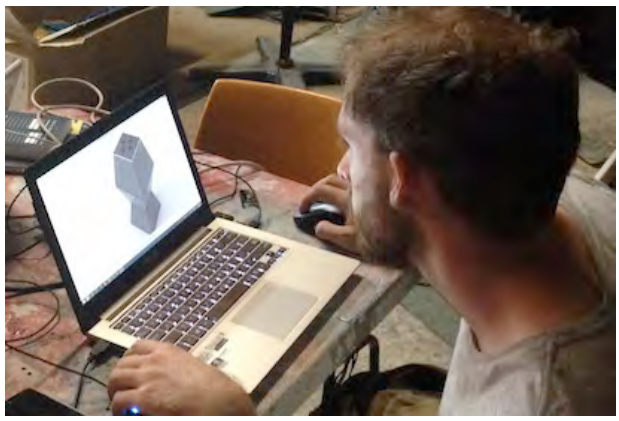


Figure 5 Developing a production design.



Figure 6 The BollART Logo.

6. INSTALLATIONS

The Sonic BollART's initial outing was during the Hack The Festival judging night[1]. Since then, it has been exhibited (or performed) at a number of events including State of the Art Festival, Come Out And Play Festival, The Totally Huge New Music Festival, the Beaufort Street Festival (an annual community street festival held in Mount Lawley, Western Australia), and TEDx Perth 2015.

[1] Footage of our presentation is available online at https://youtu.be/y83jF9PSVx8



Figure 7 The Sonic BollART deployed in the Perth Cultural Centre.

7. THE TEAM

Our Hack the Festival team consisted of: Tim Gilchrist, SKoT McDonald, Morgan Strong, Meg Travers and Craig Wale. We have since been joined by Tara Surry, as we take the project further.

Team members brought a mix of skills to the hack event including: in-depth musical knowledge, engineering, software development, documentation, 3D design, documentation and "making"; all of which contributed to our successful outcome. The maker ethos, in particular, drove our success. A useful description for understanding this ethos is summarised as "The Cult of Done" (Pettis 2009).

8. CONCLUSIONS

The constraints imposed by the art-hack were key in shaping The Sonic BollART. Developing a project during a hack event translates as coming in with broad visions[1], then simultaneously developing concrete ideas; refining them; and ultimately delivering the result. In the case of an art hack, artistic statements and rationalisation are also developed through the same process.

Being forced to rapidly develop our ideas and see them through to an implementation in a short time meant that we were not able to approach it with a clear artistic statement in mind. This had to evolve with the project, and is something which we plan to address as we take our project further.

Applying the maker ethos of "getting it done" did not allow us to overthink the concept, but did encourage us to find creative solutions to obstacles - technical, artistic or otherwise. This provided a degree of focus to our efforts, which proved positive for delivering The Sonic BollART as the winning entry for Hack the Festival 2015.

9. FUTURE WORK

Interest in The Sonic BollART project continues. Our group also has plans to bring the art vessel aspect of the BollART project to production, and have submitted a proposal to the Town of Victoria Park - in response to the Town's call for a long running public digital artwork.

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[1] at least one per team member.